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Detection of some anaemia types in human blood smears using neural networks

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Abstract

The identification process based on measuring the level of haemoglobin and the classification of red blood cells using microscopic examination of blood smears is the principal way to diagnose anaemia. This paper presents a proposed algorithm for detecting some anaemia types like sickle and elliptocytosis and trying to count them with healthy ones in human red blood smears based on the circular Hough transform and some morphological tools. Some cells with unknown shapes (not platelets or white cells) also have been detected. The extracted data from the detection process has been analyzed by neural network. The experimental results have demonstrated high accuracy, and the proposed algorithm has achieved the highest detection of around 98.9% out of all the cells in 27 microscopic images. Effectiveness rates up to 100%, 98%, and 99.3% have been achieved by using neural networks for sickle, elliptocytosis and cells with unknown shapes, respectively.

Keywords: healthy/unhealthy RBC detection and counting, circular Hough transforms, watershed segmentation, sickle cells, elliptocytosis, neural network

(Some figures may appear in colour only in the online journal)

1. Introduction

The human blood generally consists of red blood cells (RBCs), white blood cells (WBCs), platelets, and plasma. The liquid part in blood is plasma that contains softened salts and proteins. RBCs constitute approximately 40% of blood volume. WBCs are smaller in volume, but larger than RBCs. The comparative particles, which are smaller than WBCs and RBCs are called platelet cells [6, 16]. Anaemia is a sort of RBC deformation, more often than not brought about by an absence of mineral iron in the blood. Iron is very important to the human body for the creation of an iron-rich protein called haemoglobin, which helps the RBCs to carry oxygen from the lungs to the rest of the body [5, 9, 26]. This ailment occurs when the blood contains a low number of RBCs or they have insufficient haemoglobin. RBCs are continually being produced inside the spongy marrow in the huge bones of the body. The renewing of the RBCs is the main job of the marrow as it continually replaces the old cells. After approximately seven days of maturation, the new RBCs are released into the bloodstream. The normal lifespan of RBCs is about 120 d. The main function of the RBCs is to carry oxygen and remove carbon dioxide (a waste product) from the body. The shape of the RBCs is like a disk, which can move easily through the blood vessels.

Elliptocytosis is one of many red-cell membrane defects, and this disorder predisposes to anaemia. Historically, in 1904, this disease had been described, and then it was recognized as a hereditary condition in 1932. Medically, it is hard to determine hereditary elliptocytosis. In the USA, it is estimated that its incidence ranges between three and five cases per 10 000, whereas there is an estimated incidence of 60–150 cases per 10 000 in Africa and the Mediterranean, and 1500–2000 cases per 10 000 in Malaysia [4, 17].

In sickle-cell anaemia, the body generates crescent-shaped RBCs. Sickle cells contain abnormal haemoglobin, which is referred to as sickle haemoglobin or haemoglobin S, which causes the cells to grow a sickle or crescent shape. Sickle cells are highly dangerous, because they have a tendency to obstruct the flow in the bloodstream in the veins in the limbs and organs. The restricted blood flow can cause pain; organ damage, and increase the likelihood of sickness. Moreover,

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(a) (b) (c) (d) (e)

Figure 1. The differences in shape among normal RBCs. (a) Normal RBCs. (b) Sickle cell. (c) Elliptocytosis. (d) Platelets. (e) Cells with unknown shapes.

the abnormal sickle cells usually die after approximately ten to twenty days. The bone marrow cannot renew the RBCs fast enough to replace the dying ones [24]. Scientists, in their labs, are constantly working hard on improving any medicine to cure sickle-cell anaemia [9, 22]. Moreover, sickle-cell anaemia is most prevalent in people whose families come from Mediterranean nations, Africa, South or Central America, particularly Panama, the Caribbean islands, Saudi Arabia, and India. The number of patients in the United States suffering from sickle-cell anaemia ranged between 70 000 and 100 000 most of whom were African–Americans. The primary means of diagnosis relies upon a blood test examination that can recognize sickle cells. Figure 1 represents the distinction in shape among normal RBCs, sickle, elliptocytosis, platelets and cells with unknown shapes that were revealed in the smear.

Actually, the danger and threat of anaemia has motivated us to work r towards improving resistance to anaemia and its diagnosis. The diagnosis begins by detecting and counting the number of unhealthy cells in the blood compared with the number of healthy cells. This paper provides a proposed algorithm to identify healthy/unhealthy blood cells (elliptocytosis, sickle cells and cells with unknown shapes) based on the detection of their shapes using the circular Hough transform (CHT), watershed segmentation, and morphological tools. In this manner, we examine the cells' information variables with a few relations among them such as area, convex Area, perimeter, eccentricity, solidity, and ratio (see equations (5) and (6)), by applying the neural network (NN) to get a decision (agreed/not) with the outcomes extricated from the detection process and the proportion of agreement [2, 10].

The remainder of the paper is organized as follows; section 2 focuses on related studies. An overview on the CHT is presented in section 3. Section 4 introduces an overview on the NN. The proposed algorithm is discussed in section 5, and the experimental results that demonstrate the effectiveness of the model are detailed in section 6. Section 7 summarizes and concludes the paper.

2. Related studies

Recently, research on blood cell detection and the determination of diseases by utilizing image processing has increased. In 2010, Hirimutugoda and Wijayarathna proposed a strategy to recognize thalassemia and malarial parasites in blood test images obtained from light-magnifying instruments and explored the likelihood of a quick and exact robotized finding of RBC issue. To assess the precision of the grouping with regard to medicinal image patterns, they prepared two back-proliferation artificial neural network (ANN) models (three and four layers) together with image investigation systems on the morphological components of RBCs. The three layers yielded the best execution with an error of $2.74545e^{-005}$ and an 86.54% right accurate rate. The three prepared layers of an ANN comprised the most recent identification classifier to identify diseases [11].

In December 2012, Das and others presented a philosophy utilizing a few strategies of machine learning for describing RBCs in anaemia with regard to microscopic images of peripheral blood smears. First, to reduce the unevenness of the foundation brightening and noise, they preprocessed peripheral blood smear images taking into account geometric mean filter and the method of grey-world assumption. At that point, a watershed segmentation strategy was applied to erythrocyte cells. Unhealthy RBCs, for example, sickle cells, echinocyte cells, teardrop cells, elliptocytosis cells, and solid cells were classified based on their morphological shape changes. They observed that when a little subset of elements utilized data pickup measures, the logistical regression classifier exhibited better in execution. They accomplished a better forecast in general accuracy by 86.87%, sensitivity 95.3%, and specificity 94.13% [5].

In January 2013, another proposition presented by Taherisadr and others, exhibited a mechanized red cell examination and grouping from created singular examples. This system was based on the shape highlights inner focal whiteness setup of red cells and their circularity with the assistance of decision logic. RBCs were classified into 12 classes to get results, i.e. diagnosis of blood issue, for example, iron-inadequacy anaemia, chronic illness anaemia, β-thalassemia quality, sickle-cell anaemia, haemoglobin C ailment, intravascular hemolysis, hereditary elliptocytosis, genetic spherocytosis and megaloblastic anaemia because of folic acid insufficiency. In this strategy, their portrayal of a few limits to characterization has high efficiency in the coordination of results. They contemplated morphological perspectives and ascertained parameters and limits to get the quantitative consequences of cell regions and diameters [23].

Minetti and others, in March 2013, presented an overview of common mistakes using the most popular methodologies in RBC research and how to avoid them. In addition, they introduced a number of standards for data comparison between the different labs and techniques. They considered flow cytometry, patch-clamp measurements, biochemical analysis, flux measurements and dynamic fluorescence imaging as well as emerging single-cell techniques, such as the use of optical

tweezers and atomic force microscopy. They concluded that, to clarify the inter-cellular variability of responses, measurements in cell suspensions should be combined with single-cell techniques such as fluorescent live-cell imaging, flow cytometry (FCM) and/or patch-clamp approaches. However, even between single-cell techniques, there are regularly differences and confusing interpretations, because cell behaviour is highly sensitive. So, considerations that lead to better harmonisation of experimental conditions are timely and relevant, especially regarding the accumulation of large amounts of data in the literature [18].

In July 2013, Mushabe and others presented an algorithm that distinguished and counted RBCs and parasites to perform a parasitemia estimation. Morphological operations and a histogram-based threshold were used to classify RBCs. They utilized boundary curvature calculations and Delaunay triangulation to the split overlapped RBCs. A Bayesian classifier with RGB pixel values was employed as an element to classify the parasites. The outcomes indicated 98.5% sensitivity and 97.2% specificity for distinguishing tainted RBCs [20].

Tomari and others, in 2014, exhibited a proposed PC-supported framework to mechanize the procedure of detection and recognizable proof of RBCs from blood smear images. This framework comprised a blend of three fundamental stages, segmentation and handling, feature extraction, and classification. They extracted RBC regions from the background using a global threshold method applied on a green channel colour image. The Otsu segmentation method was applied on the green colour channel image with a progressive post-handling filter to concentrate the strong RBC shapes from the background. Also, the filter and connected component labelling were used to remove noise and holes. According to the feature extraction, the fusion of compactness and HU (defined as a derived set of invariants using algebraic invariants) moment invariant were successful in distinguishing the normal and abnormal shape with low computational resources and high discrimination power. At last, the ordinary/irregular cells were arranged to utilize the NN with four hidden layers. The system tried to classify the RBCs, and the outcomes accomplished 83% accuracy, 82% normal accuracy, and 76% normal review [25].

In 2014, Rashmi Mukherjee displayed an assessment of morphometric elements of placental villi and vessels in preeclampsia and typical placentae. The study included light microscopic images of placental tissue areas of 40 preeclampsia and 35 normotensive pregnant women. The villi and vessels were depicted from preprocessing and segmentation of these images. She applied principal component analysis (PCA), Fisher's linear discriminant analysis (FLDA), and hierarchical cluster analysis (HCA) to recognize placental (morphometric) highlights, which are basic features in microscopic images. She accomplished five huge morphometric highlights (>90% general discrimination accuracy) recognized by FLDA, and the PCA returned three significant principal components that cumulatively clarified 98.4% of the total variance [19].

Lee and Chen introduced, in 2014, an NN model with a classifier, which uses the visual data separated from the images of the RBC to determine if an RBC is normal or abnormal. They clustered the visual elements into two primary gatherings, in particular texture cluster groups and shape cluster groups, depending on the element properties. The input feature clusters were handled utilizing parallel and course construction with multiple input layers. Their trial results indicated a noteworthy change in order precision in the proposed framework when contrasted with the single-input-layer classifier with late feature selection algorithms [17].

In 2015, Yi and others proposed a 3D order technique for naturally deciding the morphologically typical RBCs in the image of numerous human RBCs that were acquired by pivot-advanced digital holographic microscopy (DHM). The RBC 3D images were recorded in the beginning by DHM and after that a computational numerical algorithm recreated the stage images of numerous RBCs. They chipped away at three common RBC shapes, stomatocyte, discocyte, and echinocyte for training and testing. Nonetheless, the irregular RBC shapes were characterized as a fourth class. Actually, they are subject to ten components extracted from every RBC after segmenting the reconstructed phase images by using a watershed transform algorithm, which includes projected surface area, average phase value, mean corpuscular haemoglobin, perimeter, mean corpuscular haemoglobin surface density, circularity, mean phase of the centre part, sphericity coefficient, elongation, and pallor. Likewise, they applied the PCA algorithm to diminish the dimension number of variables and built up the Gaussian mixture densities utilizing the anticipated information with the initial eight principal components. Thus, the Gaussian mixtures were used to plan the discriminant capacities in light of the Bayesian decision hypothesis. Their exploratory results demonstrated that the proposed strategy could deliver good results for establishing the rate of each ordinary typical RBC shape in a recreated stage image of various RBCs [27].

3. Overview on the CHT

Hough transform is a popular feature-extraction technique that transforms an image from its Cartesian to Polar coordinates. Hough introduced it in 1962, whereas Duda and Hart, in 1972, first used it to find lines in images and proposed an idea of transformation figure point to a cone in 3D parameter space [7, 12]. In the Hough transform, any point within the image space can be represented by a sinusoidal curve in the Hough space. Moreover, two points in a line segment generate two curves, which overlaid at a location corresponds with a line through the image space. Even though this model form is very easy, it is extremely complicated in the case of compound shapes due to incomplete shapes and noise with the problem of finding the slopes of vertical lines [8].

The detection of circles/ellipses using the CHT is more difficult than the line-detection algorithm due to the large number of parameters involved in describing the shapes. It is used to determine the parameters of a circle based on a known number of points that fall on the circumference. The parametric equations of a circle with radius r and centre (a, b) can be described as in equation (1):

$$x = a + r\cos(\theta),\ y = b + r\sin(\theta) \tag{1}$$

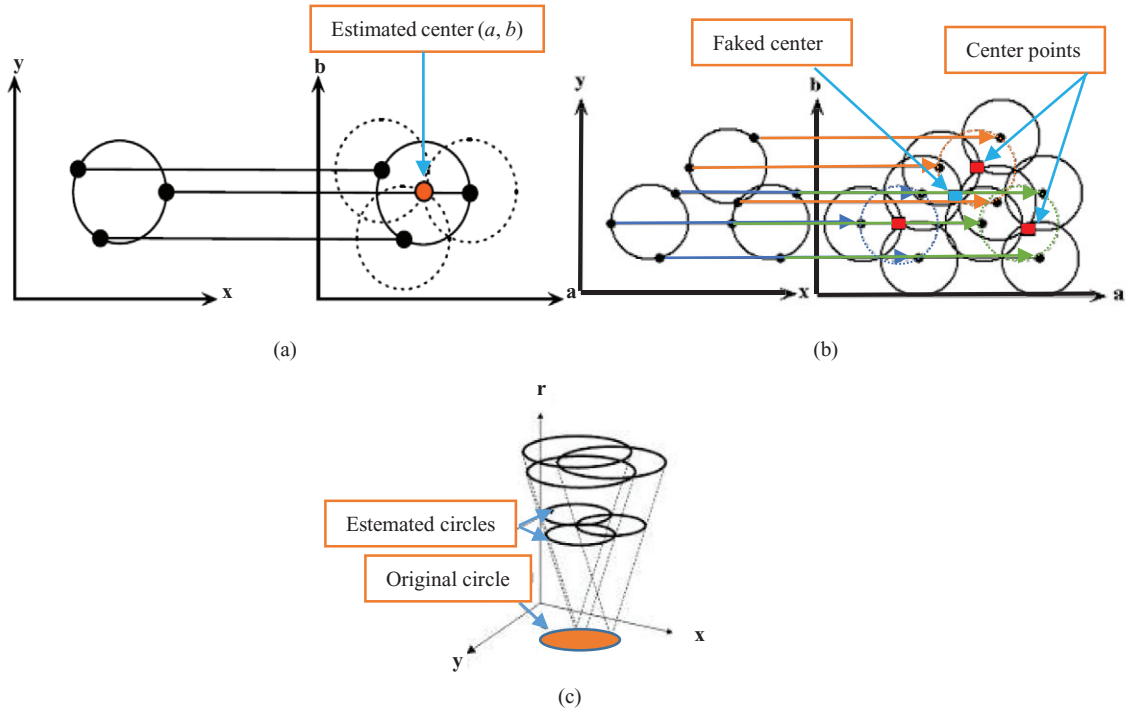


Figure 2. An example of CHT. (a) Pixels in geometric and parameter spaces. (b) Multiple and overlapped circles with known radii. (c) A conical surface in parameter space for (x, y) points on a circle.

when the angle θ measured from zero to 360 degrees, the points (x, y) trace the perimeter of a circle. The job of the estimation system is to find parameter triplets (a, b, r) to describe each circle if an image contains many points and some of them fall on perimeters of circles. The location of (a, b) points in the parameter space fall on a circle of radius r centred at (x, y). The true centre point will be common to all parameter circles, and can be found with a Hough accumulation array.

Actually, three essential steps are required for it to be a CHT. One is accumulator array computation, with its cells constructed by those candidate pixels that have high gradients and are allowed to be part of the 'votes' process. Centre estimation is the second one that estimates by detecting the peaks in the accumulator array. Finally, the radius estimation step that uses the same accumulator array [13, 14]. Figure 2(a) shows that each point in geometric space (solid dots) on the left generates a circle in parameter space (dashed circles) on the right. The circles in parameter space intersect at the (a, b) that is the centre in geometric space. In (b), the multiple circles with the same radius can be found with the same technique. The centre points, which are created by the intersection of the circles, are represented as red points in the parameter space. The overlapped circles can create fake centres, such as at the blue point. These fake circles can be removed by matching with the circles in the original image. In the same context, if the radius is not known, as in RBCs, then the location of points in parameter space will fall on the surface of a cone. Each point (x, y) on the perimeter of a circle produces a cone surface in parameter space. The triplet (a, b, r) will correspond to the accumulation cell in the array, where the largest number of cone surfaces intersect, as shown in figure 2(c). A circle with a different radius has constructed at each level r.

The estimated circles with unknown radius can be conducted by using a 3D accumulation matrix. The radii have been estimated clearly by using radial histograms, simply decoding phase information from the estimated centre located within an accumulator array.

4. Overview of NN

An NN is commonly known as a powerful function approximation for prediction and classification problems. Historically, this field started when neurophysiologist Warren McCulloch and mathematician Walter Pitts introduced a paper on how neurons might work, in 1943. They found a model for a simple NN, using electrical circuits [15]. They named this model 'threshold logic'. The model encouraged research in NNs using two distinct approaches. One concentrated on biological processes in the brain, and the other focused on the application of NNs in artificial intelligence.

In multilayer perceptron architecture, the input layer consists of the input vector $x = \{x_1, x_2, x_3, \ldots, x_k\}$. The hidden layer consists of a vector of n neurons $h = \{h_1, h_2, h_3, \ldots, h_n\}$. Finally, there is an output layer with one neuron for every element of the output vector $y = \{y_1, y_2, y_3, \ldots, y_m\}$. Every element in the input layer is connected to every neuron in the hidden layer with w_{ki} indicating the weight associated with the connection between the kth input element and the ith hidden neuron. The same connection structure is present between the hidden and output layers with w'_{ij} indicating the weight associated with the connection between the ith hidden neuron and the jth output neuron, as shown in figure 3.

In other words, weight w_{ki} can be represented as (k, i)th entry in a $K \times N$ weight matrix W and similarly weight w'_{ij} as

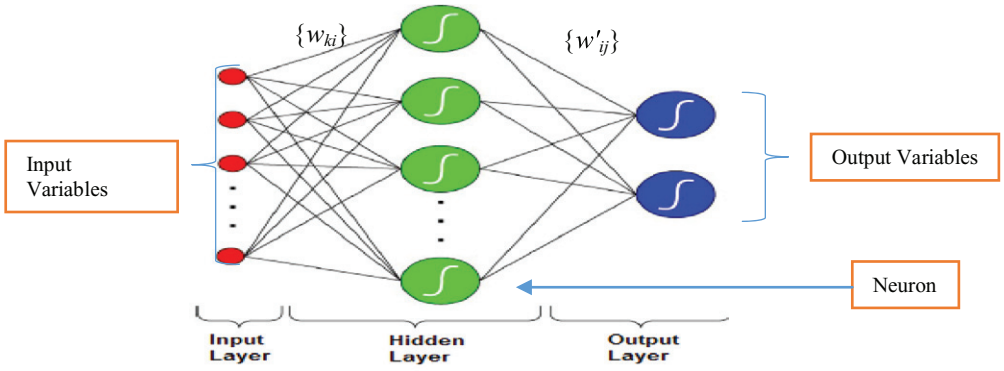


Figure 3. The structure of NN.

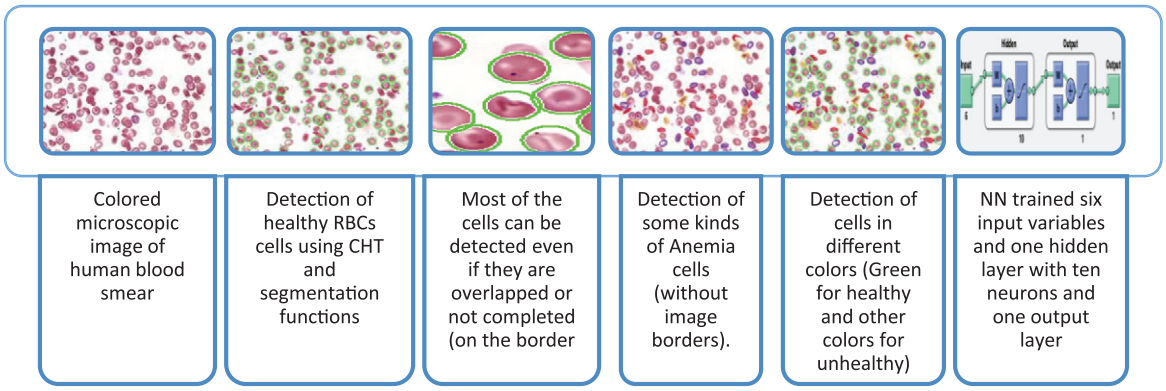


Figure 4. The main points of the proposed algorithm.

the (i, j)th entry in an $N \times M$ weight matrix W'. It is simply the logistical function applied to the weighted sum of the neuron's inputs. For example, the output of an arbitrary neuron in the hidden layer h_i is:

$$h_i = f\left(\sum_{k=1}^{K} w_{ki} x_k\right) \tag{2}$$

and similarly for the output of an arbitrary output neuron y_j is:

$$y_j = f\left(\sum_{i=1}^{N} w'_{ij} h_i\right) \tag{3}$$

Clearly, each neuron in the hidden layer is a function that maps an input vector x to scalar output vector y through weight vector w and nonlinear activation function f, which provides the nonlinearity between the input and output as in equations (2) and (3). It takes a weighted sum of the inputs and returns to y. This activation function ranges from a simple threshold function, sigmoid function, and hyperbolic tangent or radial basis function. The objective function E is summed over all elements in the output layer. It measures the difference between the actual output t_j and the predicted output y_j.

$$E = \frac{1}{2} \sum_{j=1}^{M} (y_j - t_j)^2 \tag{4}$$

The back-propagation training process is to perform a particular function by adjusting the values of the connections (weights) between elements. Actually, the dataset is normally divided into training and test subsets. The back propagation works by presenting each input sample to the network where the estimated output can be computed by performing weighted sums and transfer functions. It applies a weight correction to reduce the difference between the network outputs and the desired ones, in other words, the NN can learn, and can thus reduce the future errors as in equation (4) [3, 11].

5. The proposed algorithm

The objective of this paper is to identify healthy and a few types of unhealthy RBCs in a hued microscopic image of human blood smear and recognize them. The proposed algorithm is represented in figure 4.

Figure 4 presents an application of the CHT with morphological functions on splendid and dull power cells to recognize and count healthy/unhealthy blood cells. CHT has been employed to identify RBCs regardless of the possibility that they have been covered up, are on image borders or overlapped. At this point, watershed and morphological functions have been utilized to upgrade and isolate overlapped cells during the segmentation process. Numerous operations have been utilized through it:

- Cell polarity; it shows whether circulating blood cells are brighter or darker than the background.
- Computation method (two-stage) has been used to ascertain the accumulator array of the CHT. It depends on interpreting spread histograms; radii are plainly applied on the assessed cell focuses alongside the image [9].

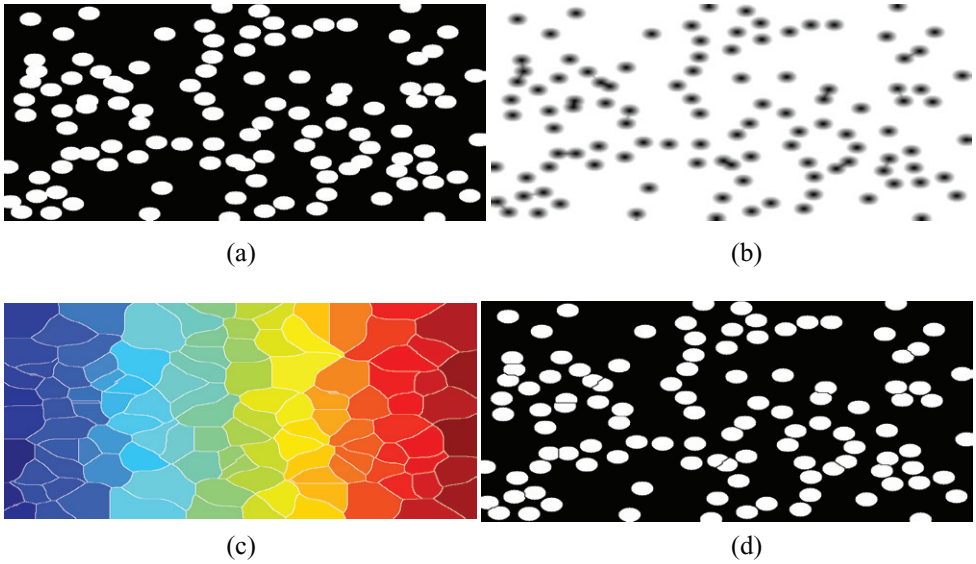


Figure 5. The watershed stages. (a) The selected healthy cells. (b) The computation of the true Euclidean distance. (c) The watershed segmentation process. (d) The new image after separation of the overlapped cells.

- Sensitivity factor is the sensor of the accumulator array in the CHT. The recognition has included defective and half covered up or overlapped cells; in any case, higher estimations of the sensitivity increase the risk of false identification.
- Edge-gradient threshold; cells have largely darker cores and are encompassed by an outlying bright halo. The edge-gradient threshold is extremely helpful for identifying edge pixels in these images of both unhealthy and strong blood cells, taking into account their well-recognized differences by setting a lower value at the threshold. It distinguishes fewer cells with defective edges by expanding the estimation of the threshold [6].

In general, image segmentation is the process of selecting objects in the image from the background, i.e. partitioning the image into disjointed regions, such that each region is homogeneous with respect to some property, such as grey value or texture. The watershed transform can be classified as a region-based segmentation approach. Simply, let f be a digital grey value image. Initially, we assume that f is lower complete, that is, each pixel, which is not at a minimum has a neighbour of lower grey value. The lower slope $\mathrm{LS}(p)$ of f at a pixel p, is defined as the maximal slope linking p to any of its neighbours of lower height, as in equation (5).

$$\mathrm{LS}(p) = \max_{q \in N_G(p) \cup \{p\}} \left(\frac{f(p) - f(q)}{d(p,q)} \right) \tag{5}$$

where $N_G(p)$ is the set of neighbours of pixel p on the grid $G = (V, E)$ of V that is the set of all vertices and E is the set of all edges, whereas $d(p, q)$ is the distance associated to edge (p, q) (for $q = p$ the expression following the max-operator is defined to be zero). In the case of pixels whose neighbours are all of higher grey value, the lower slope is zero. The cost of moving from pixel p to a neighbouring pixel q is defined as:

$$\mathrm{cost}(p,q) = \begin{cases} \mathrm{LS}(p) \cdot d(p,q) & \text{if } f(p) > f(q) \\ \mathrm{LS}(q) \cdot d(p,q) & \text{if } f(p) < f(q) \\ \frac{1}{2}(\mathrm{LS}(p) + \mathrm{LS}(q)) \cdot d(p,q) & \text{if } f(p) = f(q) \end{cases} \tag{6}$$

The distance $d(p, q)$, in equations (5) and (6), between non-neighbouring pixels p and q is defined as the minimum path length among all paths from p to q (this depends on the graph structure of the grid, in other words, the connectivity) [21]. In the proposed algorithm, the watershed segmentation has been used to help in the separation of all overlapped healthy/unhealthy cells. Figure 5 shows an example of the stages of using watershed in healthy cells. In figure 5(a), the healthy cells are filtered and selected in the grey image by using some of the morphological tools, whereas (b) gives the complementary image with some computations to get the Euclidian distance among the cells that will be used in the cost function of equation (6). In (c), the final image of watershed segmentation after distinguishing the regions has appeared, then (d) shows the final image of healthy cells after separating the overlapped cells. Therefore, all separated cells are ready and available to contour them and get their measures like radii, area, and so on, as shown in figure 4. These measures will be used later in the NN.

Figure 6 illustrates that the healthy cells have been detected using the CHT, some of the morphological functions and watershed segmentation to detect all individual cells. Then the radii and their centres and regions are measured and determined to construct all variables of the healthy cells. The next step is to apply a filter to remove all the healthy cells to get the unhealthy ones. By repeating all the previous steps with the remaining unhealthy cells, the variables concerned are measured. Therefore, the variables and targets for all healthy and unhealthy cells have been constructed. In the end, the NN has been applied to all data extracted from all variables of the healthy and unhealthy cells.



Figure 6. The separate steps of the proposed algorithm.

Table 1. Description of variables.

#	Variables	Description
1	Area	It is the actual number of pixels in the region.
2	Convex area	It is the specified number of pixels in the convex hull, with all pixels within the hull filled in (i.e. set to on). Moreover, the convex hull is an $m \times 2$ matrix that specifies the smallest convex polygon that can contain the region. Each row of the matrix contains the x- and y-coordinates of one vertex of the polygon.
3	Perimeter	It is the distance around the boundary of the region. It has calculated the distance between each adjoining pair of pixels around the border of the region.
4	Eccentricity	The eccentricity of the ellipse that has the same second-moments as the region. The eccentricity is the ratio of the distance between the foci of the ellipse and its major axis length. The value is inbetween 0 and 1. (0 and 1 are degenerate cases; an ellipse whose eccentricity is 0 is actually a circle, while an ellipse whose eccentricity is 1 is a line segment).
5	Solidity	It is the relationship or proportion of the pixels in the convex hull that are also in the region, as in equation (5).
6	Ratio	It is the relationship between area and perimeter, as in equation (6).

The NN has been applied to test and examine the execution of the proposed algorithm for diagnosing and deciding whether the patient has one or more types of anaemia. The characterization relies upon the distinguished cell information variables; area, convex area, perimeter, eccentricity, solidity, and ratio (see equations (5) and (6)) to reduce errors in the detection process and ensure that anaemia presence is established. Table 1 depicts these variables briefly.

The last point of the proposed algorithm includes the input image with all the contoured healthy (green) and unhealthy (different colours) RBCs, in which all tested images are set to the same distance (focal length) from the magnifying microscope optics to control all measures. Through this process, the information variables for every cell are measured, counted, and analyzed for the majority of types in a smear. These measures serve as the data variables to train the NN, whereas the output targets are measured with regard to the solidity (S_C) and ratio (R_C). The target solidity (S_C) is obtained by dividing the area by its convex area for all separate cells as shown in the equation (7) whereas the target ratio (R_C) is the ratio calculated by dividing the area of cells to perimeter of cells, as shown in equation (8):

$$S_C = \frac{\text{area}_c}{(\text{convex area}_c)} \tag{7}$$

where area_c is the area of each for all kinds of detected cells, and convex area_c represents the convex area of each cell for all kinds of detected healthy/unhealthy cells.

$$R_C = \frac{4\pi(\text{area}_c)}{(\text{perimeter}_c)^2} \tag{8}$$

The perimeter_c represents the perimeter of each cell for all kinds of detected healthy/unhealthy cells. The target T_{S_c} is an array that consists of binary numbers (1) and (0) for the conditions of solidity and eccentricity to classify sickle-cell anaemia, as shown in equation (9).

$$T_{S_c} = \begin{cases} 1 & \text{if } S_C < 0.72 \quad \text{OR} \quad \text{Eccentricity} > 0.9 \\ 0 & \text{if Otherwise} \end{cases} \tag{9}$$

In equation (8), if any of the solidity values S_C are less than 0.72 or the eccentricity is greater than 0.9 then it is a sickle cell. Otherwise, T_{S_c} assumes a value of zero, which decides if it is a healthy/unhealthy cell; however, the type is not a sickle-cell anaemia. This is also applicable to target T_{R_C}, which is

Table 2. Blood cell variables.

#	Variables	Type	Domain
1	Convex area	Ranged	178 : 1145
2	Perimeter	Ranged	51.1127 : 150.3675
3	Eccentricity	Ranged	0 : 0.97 553
4	Areas	Ranged	158 : 869
5	Solidity	Ranged	0.506 494 : 0.979 14
6	Ratio	Ranged	0.278 963 : 0.945 956
7	T_{S_C}	Binary	(1 for sickle cell, 0 for other healthy/unhealthy cells)
8	T_{R_C}	Binary	(1 for elliptocytosis, 0 for other healthy/unhealthy cells)
9	U_C	Binary	(1 for unknown shapes, 0 for other healthy/unhealthy cells)

an array that consists of binary numbers (1) and (0) under the conditions of ratio to classify elliptocytosis, as shown in equation (10).

$$T_{R_C} = \begin{cases} 1 & \text{if} \quad 0.7 \leqslant R_C \leqslant 0.9 \\ 0 & \text{Otherwise} \end{cases} \tag{10}$$

when T_{R_C} is 1 the ratio R_C ranges between 0.7 and 0.9, which indicates that the unhealthy cell belongs to the elliptocytosis type. When the R_C value is less than 0.7 or greater than 0.9 then T_{R_C} takes the value 0 and leads us to the conclusion that the cell is not from elliptocytosis type and may be healthy or be some other type of unhealthy cell. In equation (11), U_C is an array that has been constructed under the conditions, reverse or complementary, of all the previous conditions in T_{S_C} and T_{R_C}.

$$U_C = \begin{cases} 1 & \text{if} \quad 0.7 > R_C \text{ AND Eccentricity} \leqslant 0.9 \text{ AND } S_C \geqslant 0.72 \\ 0 & \text{Otherwise} \end{cases} \tag{11}$$

The target U_C of unknown shapes is defined by 1 if the values of R_C are less than 0.7, logically AND eccentricity is less than or equal to 0.9, also logically AND S_C is greater than or equal to 0.72; but it takes the value 0 when the value satisfies the conditions of T_{S_C} and T_{R_C} and then we conclude that it is either of the healthy, sickle, or elliptocytosis types. The back-propagation NN has trained and tested six variables as an input layer and ten neurons in the hidden layer, but one neuron in the output layer. Moreover, three kinds of samples are applied; training, validation, and testing samples. The training samples have been presented in the network during the training process by 70% of all samples in the input variables, and the network has been modified according to its error. Accordingly, only 15% has been used for the validation to measure network generalization and to pause training when generalization stops improving.The remaining 15% of all samples of cells has been introduced to be a testing sample that has no effect on training and so provides an independent measure of network performance during and after training. In addition, the mean square error (MSE) has been applied, that is defined as the average squared difference between outputs and targets, whereas lower values are better; zero means no error. Table 2 illustrates the description of the formed healthy/unhealthy cell data.

All variables have been arranged according to the same type, whereas only the output (T_{S_C}, T_{R_C}, and U_C) variable has a binary value, such as 1, to exactly defined the cells and 0 value for the others. As a final point, the execution of every classification model has been assessed utilizing three factual measures; classification accuracy, sensitivity, and specificity. These norms are set as true positive (TP), true negative (TN), false positive (FP) and false negative (FN). A TP choice is made when the positive forecast of the classifier agrees with a positive expectation of the previous segmentation. A TN choice is made when both the classifier and the segmentation propose the non-attendance of a positive expectation. An FP choice is made when the framework indicates that the cell (positive forecast) is undesirable. Finally, an FN choice is made when the framework indicates that a negative (unhealthy) cell is positive. In addition, the classification accuracy has been characterized as a proportion of accurately classified cells and is equivalent to the entirety of TP and TN separated by the aggregate number of RBCs (N) [1].

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{N} \tag{12}$$

The sensitivity alludes to the rate of accurately arranged positive cells and is equivalent to TP partitioned by the total of TP and FN.

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

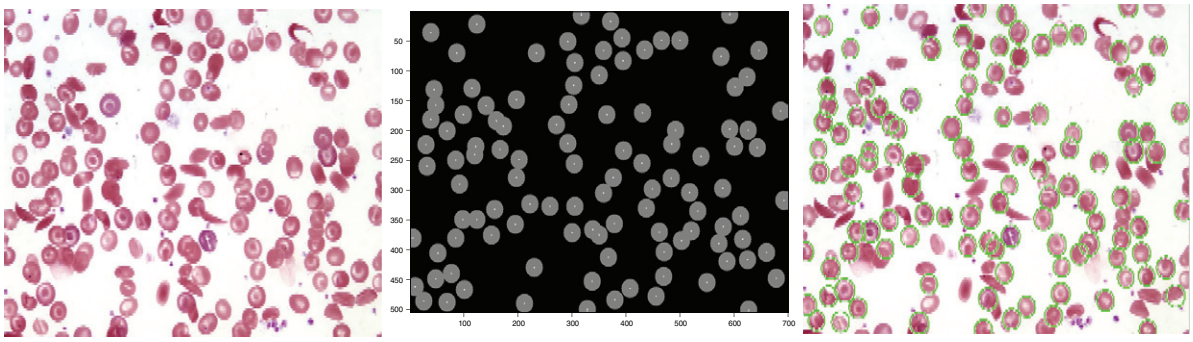
Specificity points to the rate of effectively characterized negative cells and is equivalent to the proportion of TN to the entirety of TN and FP [1].

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{14}$$

6. The experimental results

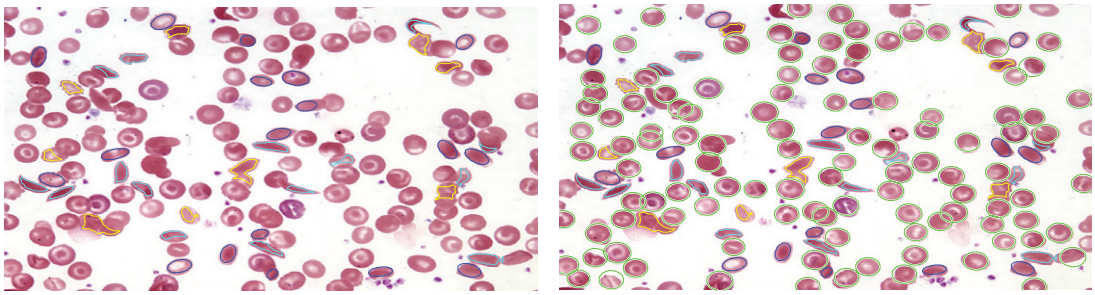
The test results are presented in two stages; one is the RBC detection and the counting of healthy/unhealthy cells. The second stage is the investigation and arranging of the selected cell information to verify the agreement with the detection process utilizing the most well-known classification model in the information examination NN [1, 2].

In the detection process, blood cells were acquired from subjects with anaemia, and the algorithm was employed using



(a) (b) (c)

Figure 7. (a) The original image (b) makes a small mark on every cell's centroid, and (c) the final detected healthy cells.



(a) (b)

Figure 8. (a) All contoured unhealthy cells, and (b) the final detection of all cells.

27 selected images with a size of 504 × 700 for ten test samples with 40 × magnification to accurately detect the anaemia type; sickle, elliptocytosis, and cells with unknown shapes. Cells were spread meagrely onto glass magnifying lens slides to permit singular cell types to be identified. Giemsa was used in blood preparation and it stained the RBCs pink, and the platelets and WBCs magenta. All cells (healthy/unhealthy) were detected using the CHT, watershed, and morphological techniques for enhancement. In the same context, the CHT was applied under the conditions of cell polarity to determine all dark and bright cells according to their intensity. Two-stage techniques are used thereafter to compute the accumulator array of the CHT; the sensitivity of this accumulator array in the proposed algorithm is 0.97 for brightness and 0.90 for dark cells, and an edge-gradient threshold 0.2 that is detected fewer cells with weak edges. Actually, these conditions help the CHT to detect most of the healthy RBCs positioned singularly, overlapped, and even attached to other unhealthy cells. Figure 7(a) shows an example image of human blood smear and some of the proposed algorithm steps are shown in (b) and (c).

According to this image, through the detection process, the number of healthy cells is 109 out of 160, which is the total number of all detected cells (healthy/unhealthy). The remaining number of cells (51) may be considered as detection errors, unhealthy cells (sickle, elliptocytosis, and cells with unknown shapes), platelets, or even WBCs. On the other hand, platelets and WBCs have been neglected (if they have appeared in the image) as the algorithm has only concentrated on healthy/unhealthy RBCs [9].

In addition, the algorithm has determined 153 of the detected blood cells, without detecting unhealthy cells on the image borders, by about 98.9% according to the image in figure 7(a). Consequently, the next step is trying to classify these 51 cells into several types of unhealthy cells. All the cells have strange shapes (crescent, elliptic, platelets, and unknown shapes): they are the detected and selected unhealthy cells according to the previous steps of healthy-cell detection. In the same way, a mask of balls is based on the cell centroids and radii (measuring 17–20 pixels in distance) to select only healthy cells from this image. After the masking process, many overlapping healthy cells appeared as one big region according to the shape and size of the normal cells. For that reason, the watershed process has been applied to get the optimum separation in these abnormal shapes. In fact, to get the optimal separation of overlapped cells their centroids must have been used to put marks on them as in figure 7(b). At the end of this process, only the healthy cells are contoured (the green line), have been extracted to be counted, and distinguished from the other unhealthy cells as shown in figure 7(c).

Figure 8 illustrates the detection of most unhealthy cells without any in the image borders. In (a) a coloured segmented image shows all the unknown unhealthy shapes with noise-like platelets, etc. The sickle cells or the cells that are identified as sickle, elliptocytosis, and cells with unknown shapes have been detected without any noise in different contour colours, and the platelets have been neglected. The blue line contours the elliptocytosis, the sickle is in red, and the yellow line represents the unknown cell shapes. In the same context, of the 51 deformed cells, 15 are identified as sickle,

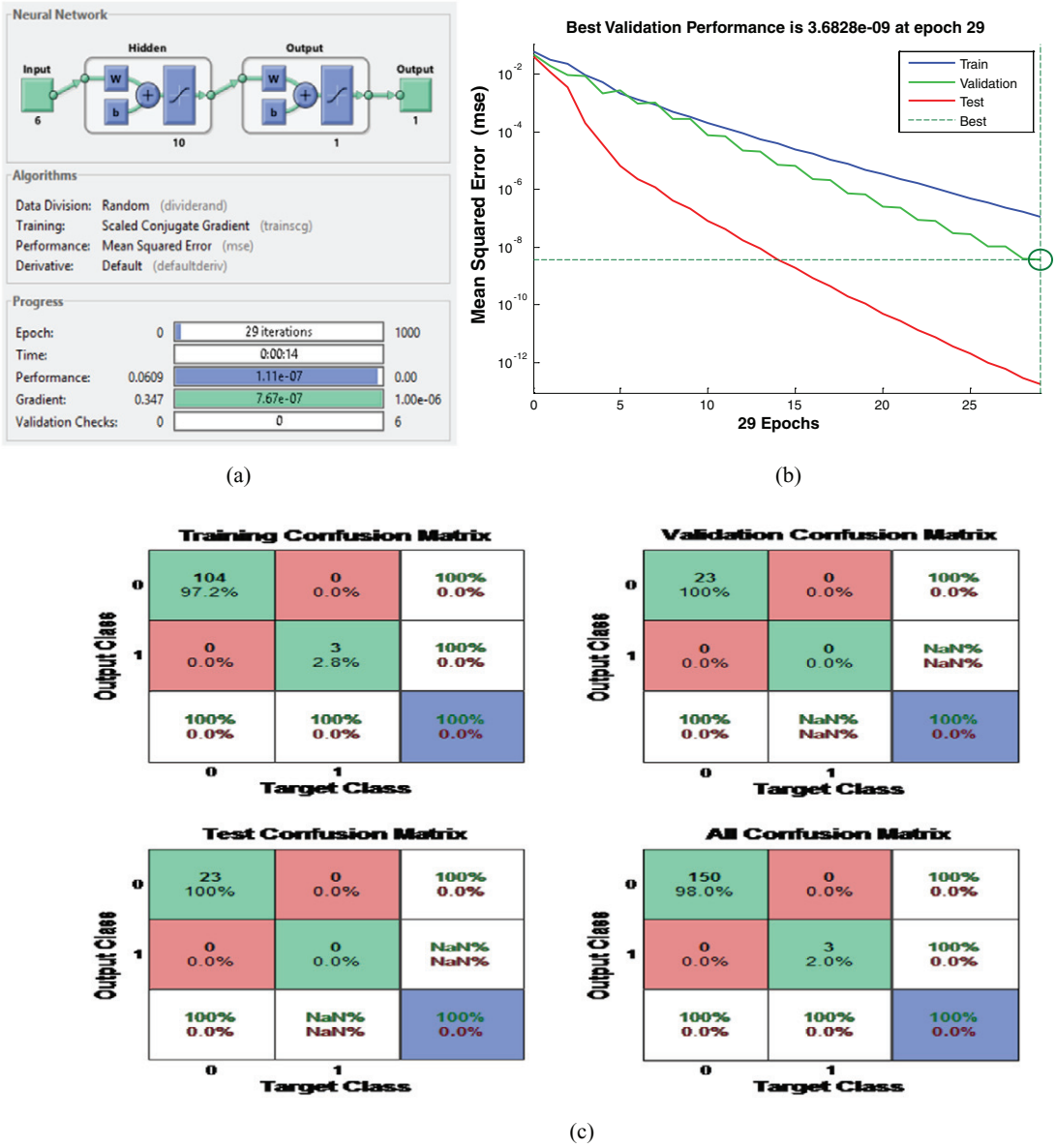


Figure 9. (a) The back propagation of the first NN sickle cells, and (b) the best validation performance, and the confusion matrices in (c).

18 are elliptocytosis, and nine are unknown cell shapes. The remaining nine unhealthy cells which are not detected may appear on the neglected image borders, as unknown detected cells of composite shapes that are not defined or known, or are perhaps waste or unknown blood disease. In figure 8 (b), the final detection of the healthy cells is shown by green lines and the unhealthy ones with a variety of coloured lines.

Experimentally, the NN consists of six input variables, ten neurons in one hidden layer and one output layer. Actually, two networks have been trained with two targets T_{Sc}, and T_{Rc} for detection of sickle, and elliptocytosis, and the third one has been trained for unknown shapes with target U_C. First, the NN is concerned with the detection of the sickle cells. It succeeded in 14 s after 29 iterations out of 1000 as maximum iterations of the epoch, performance $1.11e^{-07}$, gradient $7.67e^{-07}$, and 0 validation checks as shown in figure 9(a). Figure 9(b) presents the best validation performance which is shown as $3.6828e^{-09}$ at epoch 29, where the training is denoted by a blue line, the validation is denoted by green, and the test by the red line. Moreover, the MSE of the training, validation, and testing processes is $1.10875e^{-07}$, $3.6828e^{-09}$, and $1.8058e^{-13}$, respectively. The confusion matrices for training, validation, and testing processes of the first NN are shown in (c). The predictions of the NN model are compared with the original classes of the target T_{Sc} to identify the values of TPs, TNs, FPs, and FN. These values have been computed to construct the confusion matrix, where each cell contains the number of cases classified for the corresponding combination of desired and actual classifier outputs. It has achieved 100%. Accordingly, accuracy, sensitivity, and specificity achieved 100%, 100%, and 100%, respectively.

In the second NN, the detection of elliptocytosis succeeded in 0 s after 56 iterations out of 1000 as maximum iterations of the epoch, performance 0.009 08, gradient 0.0643, and six validation checks as in figure 10(a). In the same context, the best validation performance is shown as 0.029 644 at epoch 50, with the training

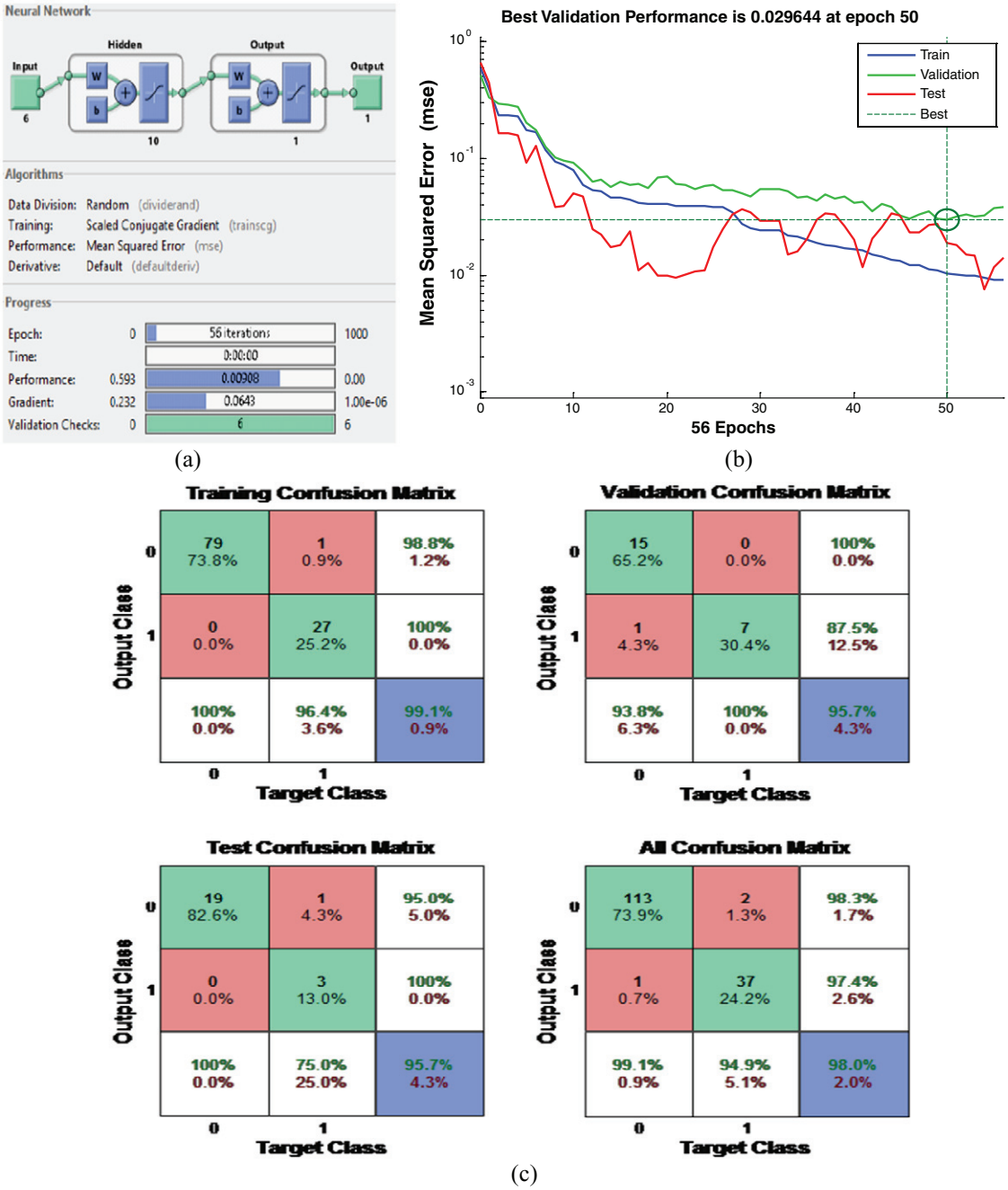


Figure 10. (a) The back propagation of the second NN elliptocytosis, (b) the best validation performance, and the confusion matrices in (c).

in a blue line, validation in green, and the test in the red line of (b). The MSE of the training, validation, and testing processes are 1.02479×10^{-2}, 2.9644×10^{-2}, and 1.87565×10^{-2}, respectively. Furthermore, the confusion matrices of the second NN with target T_{R_C} have been displayed in (c), and achieved 98%. Accordingly, accuracy, sensitivity, and specificity have achieved 99.1%, 96.4%, and 100%, respectively, success of the training samples. On the other hand, they have achieved 95.7%, 75%, and 100%, success of the testing samples.

Finally, the third NN is concerned with the detection of the unknown shaped cells. It succeeded, in 15 s, after 39 iterations out of 1000 as maximum iterations of the epoch, performance 0.000 371, gradient 0.002 29, and six validation checks as in figure 11(a).

Consequently, the best validation performance is 0.005 3972 at epoch 33, with the same colour lines as shown in (b). The MSE of the training, validation, and testing processes are $1.60196e^{-03}$, $5.3972e^{-03}$, and $1.74046e^{-02}$, respectively. The confusion matrices of the third NN with target U_C are shown in (c), and achieved 99.3%. Accuracy, sensitivity, and specificity have achieved 100%, 100%, and 100%, respectively, success of the training samples; but they have achieved 95.7%, 100%, and 95.5%, success of the testing samples.

Finally, the proposed algorithm is very effective in diagnosing, counting, and detecting healthy/unhealthy cells of a human blood smear. In the end, Matlab-2013a has been used to build the algorithm on Windows 10 with processor Intel® Core™2Duo CPU T5550@ 1.83 GHz and 2.50 GB RAM

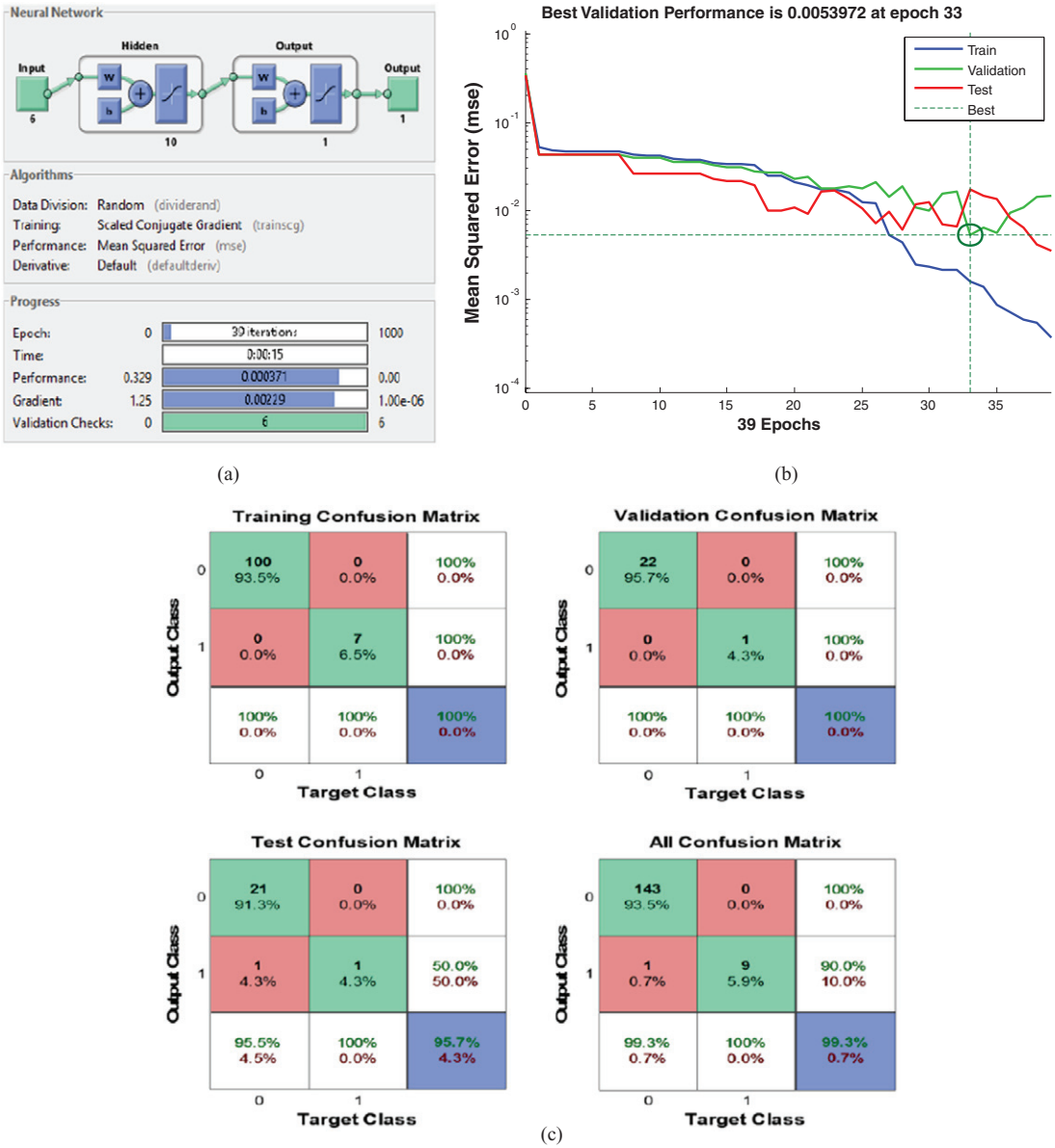


Figure 11. (a) The back propagation of the third NN of unknown shapes, (b) the best validation performance, and the confusion matrices in (c).

with a 32-bit operating system. An optical Nikon microscope has digitized all the images of the blood smear cells.

7. Conclusions

The diagnosing of healthy/unhealthy cells based on microscopic image analysis of human blood smear has become an important tool. Sickle and elliptocytosis are the most important and common types of anaemia. This paper has proposed an algorithm that can detect sickle, elliptocytosis and cells with unknown shapes and healthy ones based on microscopic blood smear. The CHT, watershed segmentation and some morphological tools have been used in the detection process. On the other hand, the extracted data have formed variables such as area, convex area, perimeter, eccentricity, solidity, and ratio and have been classified as input variables in NNs. In the same context, the functions T_{S_C}, T_{R_C} and U_C have been specified as targets in three NNs to analyze sickle, elliptocytosis and cells with unknown shapes, respectively. The performance has been calculated by three statistical measures; classification accuracy, sensitivity, and specificity. This algorithm has been successful for about 98.9% out of all input cells, which may have contributed to the improvement in the diagnostic process. The results indicate that an effectiveness of 100%, 98%, and 99.3% in the NNs, has been achieved for sickle, elliptocytosis and cells with unknown shapes, respectively. Finally, the proposed algorithm is provides an effective means for diagnosing and counting healthy/unhealthy red blood smear cells based on medical image processing.

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